

BISMARCK, ND, USAWMO: **727640**Lat: **46.783N**Lon: **100.757W**Elev: **1651**StdP: **13.84**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **24011****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-11.8	-24.1	1.5	-16.8	-18.4	2.1	-11.2	30.2	19.7	27.0	16.6	6.7	320	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.9	93.1	70.3	89.4	69.3	86.1	68.0	74.7	86.1	72.4	84.1	70.4	82.4	12.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	122.0	82.0	68.6	111.8	79.0	66.4	103.3	76.2	39.4	85.5	37.2	84.0	35.3	82.6	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(4) 26.9	24.0	20.6	-26.6	101.1	6.5	4.5	-31.2	104.4	-35.1	107.0	-38.7	109.5	-43.4	112.8
(5)			-25.8	77.9	5.1	2.4	-29.5	79.7	-32.5	81.1	-35.3	82.5	-39.0	84.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	13.0	17.1	29.6	43.4	55.4	65.6	71.9	69.8	60.3	45.5	30.4	17.9
	DBStd	23.07	14.53	13.82	13.51	10.25	8.42	6.86	5.93	6.29	8.49	9.17	11.26	12.81
	HDD50	4784	1148	922	636	241	43	1	0	0	13	196	589	995
	HDD65	8414	1613	1342	1096	648	314	74	11	25	188	606	1037	1460
	CDD50	2401	0	0	5	44	210	468	680	614	322	56	2	0
	CDD65	555	0	0	0	1	15	91	226	174	47	1	0	0
	CDH74	6756	0	0	2	46	287	1074	2561	2033	689	64	0	0
	CDH80	2636	0	0	0	9	71	333	1096	861	251	15	0	0
	WSAvg	9.3	8.9	9.1	10.0	10.7	10.5	9.3	8.4	8.4	8.9	9.5	8.8	8.7
	PrecAvg	17.1	0.5	0.5	0.8	1.5	2.4	3.1	2.6	2.1	1.5	1.1	0.6	0.6
(15) Precipitation	PrecMax	27.0	2.2	1.7	3.2	5.5	7.4	8.0	13.7	9.3	6.9	4.7	2.4	2.1
	PrecMin	10.2	0.0	0.0	0.1	0.1	0.2	0.5	0.2	0.0	0.0	0.0	0.0	0.0
	PrecStd	4.4	0.4	0.4	0.6	1.1	1.7	1.6	2.1	1.8	1.3	1.1	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	47.8	52.7	70.5	80.0	86.9	92.1	99.8	98.6	93.4	81.4	65.9	50.2	
	0.4% MCWB	39.6	41.7	51.5	57.4	64.7	71.0	71.0	71.3	68.0	60.2	51.0	40.5	
	2% DB	41.5	44.8	61.4	72.5	81.2	87.3	93.6	92.6	86.6	72.9	57.5	44.0	
	2% MCWB	36.2	37.8	47.8	54.1	62.3	69.1	71.4	69.4	65.8	56.4	45.5	37.3	
	5% DB	37.5	39.6	54.1	67.2	76.6	83.7	89.6	88.6	81.9	67.0	51.6	39.1	
	5% MCWB	33.8	35.0	44.2	50.8	59.4	66.7	70.5	68.6	63.5	53.0	42.3	34.2	
	10% DB	33.7	35.9	48.0	62.0	72.2	80.3	86.1	84.7	77.0	61.2	46.0	34.9	
	10% MCWB	31.0	32.7	40.7	48.1	57.3	64.6	69.2	67.2	61.3	50.5	39.1	31.2	
	0.4% WB	39.8	42.5	53.4	59.4	68.3	76.0	78.4	76.9	71.4	62.7	51.4	41.3	
	0.4% MCDB	47.2	51.5	67.1	75.9	81.5	87.1	88.7	87.6	85.1	76.1	64.0	48.9	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	36.4	38.3	48.5	54.9	65.1	72.2	75.5	73.3	68.7	58.3	46.4	37.6	
	2% MCDB	40.9	43.8	59.9	70.2	77.5	82.7	87.0	84.8	81.0	69.4	56.4	43.6	
	5% WB	33.9	35.3	44.4	51.9	61.7	69.2	73.3	71.0	66.1	54.5	42.8	34.4	
	5% MCDB	37.5	39.1	53.7	64.7	72.8	80.8	84.7	84.1	77.1	65.0	50.9	38.9	
	10% WB	31.0	32.9	40.8	49.2	59.0	66.8	71.2	69.0	63.3	51.3	39.4	31.4	
	10% MCDB	33.7	35.9	47.6	60.7	69.4	77.5	82.6	81.4	74.5	60.2	45.7	34.8	
(35) Mean Daily Temperature Range	MDBR	18.9	19.1	20.3	23.7	23.8	22.8	24.9	25.8	25.4	22.3	20.1	17.8	
	5% DB MCDBR	22.9	23.4	31.3	34.1	31.1	28.5	30.8	32.0	33.6	34.3	29.9	24.3	
	5% DB MCWBR	18.4	17.5	19.3	18.2	15.4	13.5	12.5	12.9	15.2	19.1	19.0	18.3	
	5% WB MCDBR	22.5	22.9	30.5	31.4	27.1	25.2	25.5	27.2	27.8	31.0	29.0	24.4	
	5% WB MCWBR	18.1	17.3	19.3	17.4	14.7	13.4	12.7	13.1	13.8	18.1	18.9	18.7	
(40) Clear-Sky Solar Irradiance	taub	0.264	0.270	0.294	0.333	0.355	0.376	0.382	0.389	0.345	0.317	0.282	0.261	
	taud	2.430	2.442	2.462	2.381	2.390	2.374	2.385	2.339	2.443	2.484	2.472	2.451	
	Ebn at Noon	269	291	298	293	288	282	278	271	276	268	258	256	
	Edh at Noon	21	26	30	36	36	37	36	37	30	25	20	18	
(44) All-Sky Solar Radiation	RadAvg	523	857	1197	1559	1826	2039	2123	1764	1351	851	535	390	
	RadStd	43	83	93	105	154	153	97	130	95	101	75	28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page